

Yufan Du

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Education

Peking University **Sep. 2021 - Present**
Bachelor of Science in Applied Physics *and* Computer Science Dual Major *Beijing, China*

Technical Skills

Proficient in: C/C++, Python

Knowledgeable with: Python ML libraries (PyTorch, NumPy), CUDA/parallel computing, Embedded systems, FPGA, SQL/Database, JavaScript/HTML, Verilog/VHDL, Linux systems, MATLAB, Tcl, L^AT_EX

EDA tools: Familiar with Cadence Innovus, Cadence Virtuoso.

Relevant Scores

Overall GPA: 3.852 (Top 5%)
TOEFL: 104
GRE: 324 (Quantitative + Verbal), 4.0 (Analytical Writing)

Project Experiences

FPGA-Based Snake Game Design Project **Mar. 2023 - Jun. 2023**

- Advisor: Professor Xiaohui Duan.
- Deployed a computer vision algorithm and the classic Snake game on a resource-limited FPGA platform, optimizing resource usage.
- Developed and implemented a real-time control mechanism for the game using hand movement detection with a camera, enhancing user interaction.
- Introduced a 3D human-computer interface to improve player engagement and provided valuable insights for IoT edge device development.

AI-Assisted Schedule Manager **May. 2023 - July. 2023**

- Advisor: Professor Wei Guo.
- Developed a cross-platform scheduling application integrated with AI, featuring NLP for user-friendly schedule management.
- Achieved a substantial improvement in user experience by enabling efficient management and reminders through conversational interfaces.

Research Experiences

Cross-Stage Power Prediction for Integrated Circuits **Jun. 2023. - Nov. 2023.**

- Advisor: Professor Yibo Lin.
- Proposed an innovative VLSI circuit power prediction framework that integrates cross-stage circuit-transformation-aware learning.
- Achieved a significant reduction in error rates and computation time compared to industry-leading commercial tools, verified using CircuitNet benchmarks.
- It thus enabled effective power optimization strategies in early-stage VLSI design.
- *Manuscript currently under review for DAC24.*

Gate-level Circuit Simulation Research

Nov. 2023 - Present

- Advisor: Professor Yibo Lin.
- Leveraged CUDA for parallel computing to accelerate gate-level circuit simulations.
- Concentrated on optimizing simulation performance for circuits with low switching activity inputs which is more common in modern VLSI circuit analysis.

Relevant Publications

Under Review:

- “PowPrediCT: Cross-Stage Power Prediction with Circuit-Transformation-Aware Learning,” submitted to *Design Automation Conference (DAC24)*.